Supporting Information

Multi-View Contrastive Representation Learning for Liquid Chromatography Retention Time Prediction

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This supplement documents the processed molecular datasets, fixed partitions, model configurations, saved-prediction results, and system-specific evaluation protocols supporting the accompanying manuscript. Retention times and absolute-error metrics are expressed in seconds unless stated otherwise.

Organization of the supplement

Section	Contents
S1	Processed datasets, cleaning, fixed partitions and molecular overlap checks
S2	CCI/PureGIN configurations, ensemble weights, selection rules and recorded environments
S3	Retained baseline records and component/RTCL ablation results
S4	Fixed-coefficient scaffold evaluation and nested PredRet adaptation results
S5	Companion CSV/JSON files and references

Data and result conventions

Each molecule is associated with a stable sample identifier. SMRT random and scaffold partitions share the same 77,980 processed records. PredRet contains 2,197 cleaned system-specific records. Companion CSV files provide the exact assignments and saved predictions; JSON files contain model arguments, protocol definitions, source hashes and verification results.

Tabulated metrics were recomputed in double precision from saved predictions and their partition labels. Model fitting and neural-network inference were not performed during preparation of this supplement. SD denotes the sample standard deviation (ddof = 1).

S1. Processed datasets and fixed partitions

S1.1. SMRT input records

The processed SMRT inputs were retained in their fixed train/validation/test assignments. The unified-data preparation routine converts the baseline split tables into a common schema: sample_id, smiles, rt and split. The SMILES and Morgan branches consume the same molecular record and target at each index. RT labels are recorded in seconds. The public SMRT resource is identified in reference S1.

Table S1. Statistics of the fixed SMRT partitions.

Partition	Subset	n	Min RT	Max RT	Mean RT	SD RT
Random	train	63163	340.0	1471.7	808.72	174.71
Random	val	7019	496.3	1453.3	808.04	172.63
Random	test	7798	365.0	1461.0	809.43	176.51
Scaffold	train	63088	340.0	1471.7	791.33	168.48
Scaffold	val	7035	524.6	1428.6	848.66	172.61
Scaffold	test	7857	442.4	1449.3	912.68	183.45

Canonical molecular identities were recomputed with RDKit 2025.09.2 using canonical isomeric SMILES. Bemis-Murcko scaffold strings were generated with includeChirality = False. The scaffold companion table includes both the stored group key and the recomputed scaffold string.

The two assignment files use the same sample_id namespace. Identifier prefixes retain the original SMRT naming; the explicit split field specifies membership in each assignment file. Structural checks use molecular identities and group keys in addition to identifiers.

Companion files: S1_smrt_random_assignments.csv; S1_smrt_scaffold_assignments.csv; S1_smrt_split_statistics.csv.

S1.2. PredRet processing and per-system statistics

The selected ten-system input contained 2,392 records. Retention times had been converted from minutes to seconds during source-data conversion. Records with nonfinite or non-positive RT were removed; records sharing the same (chromatographic system, SMILES) key were merged using their median RT. The positive-RT input contained 2,391 records; merging collapsed 194 duplicate records to yield 2,197 cleaned records. The cleaned table retains the merge count and source RT statistics. PredRet source information is given in reference S2.

Table S2. Per-system cleaning balance and RT range.

System	Input	Removed	Collapsed	Cleaned	Min RT	Max RT
Cao_HILIC	621	0	101	520	24.06	1095.60
Eawag_XBridgeC18	364	0	0	364	48.00	1224.00
FEM_Lipids	101	1	14	86	72.00	1494.00
FEM_long	420	0	8	412	58.80	3672.00
FEM_short	136	0	0	136	74.55	1804.20
IPB_Halle	82	0	6	76	24.44	719.80
LIFE_new	184	0	11	173	24.21	330.87
LIFE_old	194	0	11	183	30.59	336.99
MTBLS87	147	0	1	146	228.00	1488.00
UniToyama_Atlantis	143	0	42	101	221.60	1920.80

Removed counts refer to nonfinite/non-positive RT records. Collapsed counts are valid input records minus median-merged records. Counts refer to system-specific observations.

Table S3. Descriptive statistics of the cleaned PredRet systems.

System	Mean RT	Median RT	SD RT	Merged groups
Cao_HILIC	251.93	182.76	231.37	81
Eawag_XBridgeC18	460.40	435.00	236.62	0
FEM_Lipids	591.39	553.68	400.91	14
FEM_long	1388.01	1434.00	1044.29	8
FEM_short	866.81	873.90	396.05	0
IPB_Halle	203.18	189.00	162.74	6
LIFE_new	124.55	97.97	90.15	11
LIFE_old	101.93	85.70	68.25	11
MTBLS87	686.05	720.00	253.22	1
UniToyama_Atlantis	948.99	918.10	315.22	23

S1.3. Molecular overlap and nested partition checks

Table S4. Shared unique identities/groups between SMRT subsets.

Partition	Subset pair	Identity/group key	Shared
random	train / val	sample id	0
random	train / val	canonical smiles	0
random	train / val	murcko scaffold	2888
random	train / test	sample id	0
random	train / test	canonical smiles	0
random	train / test	murcko scaffold	3172
random	val / test	sample id	0
random	val / test	canonical smiles	0
random	val / test	murcko scaffold	1350
scaffold	train / val	sample id	0
scaffold	train / val	canonical smiles	0
scaffold	train / val	murcko scaffold	0
scaffold	train / test	sample id	0
scaffold	train / test	canonical smiles	0
scaffold	train / test	murcko scaffold	0
scaffold	val / test	sample id	0
scaffold	val / test	canonical smiles	0
scaffold	val / test	murcko scaffold	0

Table S5. Canonical molecular identities shared with cleaned PredRet.

SMRT subset	PredRet comparison	Shared identities
train	All ten cleaned systems	2
val	All ten cleaned systems	0
test	All ten cleaned systems	3

Record-level matches are supplied in S1_predret_smrt_shared_identities.csv. Pairwise identity counts between PredRet systems are supplied in S1_overlap_checks.csv.

For each PredRet system, GroupKFold assigned ten outer folds using canonical SMILES as the grouping key. Approximately 10% of each outer-training portion was reserved for inner validation (seed 20260909). All 100 system-fold combinations had zero pairwise sample-ID and canonical-SMILES overlap among inner train, inner validation and outer test. Each cleaned record appeared once as outer test within its own system.

S2. CCI architecture and training configuration

Table S6. Common CCI architecture and input settings.

Component	Setting
SMILES backbone	ChemBERTa-zinc-base-v1; six-layer RoBERTa; hidden states from layers 2, 4 and 6
Sequence length	Maximum 256 tokens; attention mask retained
SMILES encoder	One bidirectional LSTM; 128 hidden units per direction; first-token pooling; output dimension 256
Morgan input	Radius 2; 2,048 bits; chirality retained; 16 bits per token; 128 fingerprint tokens plus CLS
Morgan Transformer	Two layers; hidden dimension 256; eight attention heads; feed-forward dimension 512
Residual cross-attention	SMILES query; Morgan key/value; self-attention, cross-attention, AddNorm and feed-forward operations; dimension 256; eight heads
Regression representation	Concatenated pooled SMILES, cross-view and Morgan representations; input dimension 768
Regression head	768 -> 256 -> 128 -> 1; GELU and dropout between fully connected layers
Contrastive projections	MLP projections to 256 dimensions, LayerNorm and L2 normalization

Table S7. Common settings recorded for the six CCI members.

Parameter	Value
Optimizer / learning rates	AdamW; encoders 0.00001, other parameters 0.0001; weight decay 0.001
Batch size / checkpoint	Train 16; validation/test 32; maximum 200 epochs; patience 20; minimum validation MAE checkpoint
Learning-rate scheduler	ReduceLROnPlateau monitored on validation MAE
Target / regression objective	Train-fitted Yeo-Johnson transformation with standardization; 0.8 MAE + 0.2 Smooth L1 (beta = 1)
RTCL objective	Rank-based negative normalized similarity ratio; temperature 0.07; omega_align = 0.01; no logarithmic transform
Prediction units	Inverse-transform regression output to seconds; one original SMILES view

S2.1. Member-specific settings and ensemble weights

Table S8. CCI checkpoint settings and inner-ensemble coefficients.

Member	Seed	Best epoch	Dropout	Aug. prob.	Fusion loss	Sim. weight	Inner weight
exact1234	1234	90	0.10	0.35	0.005	0.05	0.30
exact2025	2025	62	0.10	0.35	0.005	0.05	0.15
exact777	777	91	0.10	0.35	0.005	0.05	0.00
dropout015_seed777	777	89	0.15	0.35	0.005	0.05	0.25
lowfusion_seed777	777	76	0.10	0.35	0.002	0.03	0.10
aug050_seed777	777	75	0.10	0.50	0.005	0.05	0.20

Member labels are stable aliases for checkpoint files in S2_cci_member_configuration.csv. Aug. prob. is the training SMILES augmentation probability. Fusion loss is omega_fusion; Sim. weight is the paired projection-similarity coefficient inside the cross-view objective. All six candidates are retained in the configuration inventory.

The CCI ensemble is the weighted sum of its six candidate predictions. The final Scheme A predictor combines that ensemble with the three PureGIN outputs:

y_final = 0.30 y_CCI,ens + 0.24 y_GIN2026 + 0.18 y_GIN3407 + 0.28 y_GIN777.

Table S9. Coefficients in the expanded final prediction.

Predictor	Within-branch weight	Outer coefficient	Final coefficient
exact1234	0.300	0.30	0.090
exact2025	0.150	0.30	0.045
exact777	0.000	0.30	0.000
dropout015_seed777	0.250	0.30	0.075
lowfusion_seed777	0.100	0.30	0.030
aug050_seed777	0.200	0.30	0.060
PureGIN_2026	1.000	0.24	0.240
PureGIN_3407	1.000	0.18	0.180
PureGIN_777	1.000	0.28	0.280

CCI checkpoints were selected by validation MAE. The adopted inner coefficients are nonnegative and sum to one; the outer coefficients also sum to one. The four source prediction definitions are the same on validation and test.

S2.2. PureGIN configuration and recorded environments

Table S10. PureGIN architecture and optimization.

Item	Setting
Graph construction	RDKit molecular graph with explicit hydrogens; atom feature projection and normalized weighted adjacency
Message passing	Five GIN-style blocks; hidden dimension 256; LayerNorm; residual updates
Readout	Mean pooling and max pooling, concatenated with a 64-dimensional projection of ten RDKit descriptors
Regression head	576 -> 256 -> 64 -> 1; ReLU; dropout 0.15
Optimizer / objective	AdamW; learning rate 0.0008; weight decay 0.0001; Smooth L1 with beta = 20
Schedule / checkpoint	CosineAnnealingWarmRestarts, T_0 = 20; maximum 180 epochs; patience 25; best validation MAE checkpoint
Batch size	128 for train, validation and test

Table S11. PureGIN checkpoints and saved-prediction metrics.

Seed	Best epoch	Val MAE	Test MAE	Test RMSE	Test R2
2026	160	30.5388	27.8499	53.9337	0.9066
3407	180	31.2548	28.6455	54.6842	0.9040
777	178	30.2154	27.5287	54.0621	0.9062

Table S12. Recorded runtime environments by operation.

Operation	Python	PyTorch	RDKit	GPU
Release verification	3.9.25	2.3.0+cu118	2022.09.5	NVIDIA GeForce RTX 4070 Laptop GPU
CCI reevaluation	3.9.25	2.3.0+cu118	2025.09.2	NVIDIA GeForce RTX 4070 Laptop GPU
Scaffold run	3.12.3	2.7.1+cu128	2025.09.2	NVIDIA GeForce RTX 4090

Full machine-readable arguments, architecture dimensions, checkpoint hashes and operation-specific versions are provided in the JSON files. The SI data-processing runtime is recorded separately in metadata/recorded_environments.json.

S3. Baseline and component result records

Retained baseline prediction tables were matched to the fixed SMRT test set by sample_id, SMILES and experimental RT before calculating the metrics below. Sources are recorded in S3_baseline_metrics_and_sources.csv; prediction values are provided as a companion table.

Table S13. Retained baseline and ensemble results on the fixed test set.

Predictor	n	MAE	RMSE	MedAE	MAPE (%)	R2
GCN	7798	34.2842	61.7343	19.1850	4.1772	0.8777
1D-CNN	7798	37.2549	63.2992	21.7080	4.6210	0.8714
GNN-RT	7798	43.7288	72.7767	25.8451	5.5945	0.8300
CCI Top6 ensemble	7798	29.8620	56.2335	16.2066	3.6701	0.8985
MCRL Scheme A	7798	25.8893	52.2262	13.6490	3.2130	0.9124

S3.1. Saved component configurations

Component-run identifiers specify the settings: A, ChemBERTa; B, ChemBERTa + BiLSTM; C, B plus Morgan with concatenation; D, residual cross-attention without RTCL; E, residual cross-attention with RTCL; G, ordinary cross-attention without RTCL. Complete retained-run arguments are supplied in configs/ablation_run_arguments.json.

Table S14. Saved component-run results, recomputed from aligned predictions.

Variant	Seed	MAE	RMSE	MedAE	R2
A	2026	34.0092	59.5432	19.6114	0.8862
B	2026	32.7605	59.2079	18.6230	0.8875
C	2026	31.9276	58.6029	18.1320	0.8898
D	2026	31.6815	57.6423	17.9441	0.8933
D	3407	31.8634	57.5745	17.8436	0.8936
D	777	31.9200	58.6201	17.7636	0.8897
E	2026	31.6605	58.1826	17.6448	0.8913
E	3407	32.5804	58.9514	18.4368	0.8884
E	777	31.3155	57.6936	17.4441	0.8932
G	2026	32.6084	59.1890	18.4756	0.8875

S3.2. Three-seed RTCL comparison

For each seed, residual cross-attention without RTCL (D) is paired with the configuration containing RTCL (E). The difference is MAE_without - MAE_with; a positive value indicates lower error for the RTCL configuration.

Table S15. Seed-paired residual cross-attention results.

Seed	Without RTCL MAE	With RTCL MAE	MAE reduction
2026	31.6815	31.6605	0.0210
777	31.9200	31.3155	0.6045
3407	31.8634	32.5804	-0.7170
Mean +/- SD	31.8216 +/- 0.1246	31.8521 +/- 0.6538	-0.0305 +/- 0.6622

The three-seed mean test MAE was 31.8216 s for D and 31.8521 s for E. Individual-seed results and prediction values are preserved in the companion files.

S3.3. Final prediction records

Table S16. CCI ensemble and final fusion on validation and test.

Predictor	Subset	n	MAE	RMSE	R2
CCI Top6	val	7019	31.9463	62.1650	0.8703
MCRL Scheme A	val	7019	28.4329	60.3321	0.8778
CCI Top6	test	7798	29.8620	56.2335	0.8985
MCRL Scheme A	test	7798	25.8893	52.2262	0.9124

The final fusion combines the CCI ensemble and all three PureGIN outputs with the coefficients in Table S9. Saved-prediction test MAE decreased from 29.8620 to 25.8893 s, an absolute reduction of 3.9727 s. Validation/test records use the same uncalibrated CCI ensemble definition.

S4. Scaffold-disjoint evaluation

The fixed scaffold partition contains 63,088 training, 7,035 validation and 7,857 test molecules. Molecular structures were grouped by Bemis-Murcko scaffold. The assignment table and Table S4 document zero sample-ID, canonical-SMILES and recomputed Murcko-scaffold intersection between the subsets.

The adopted prediction uses five active CCI members and three scaffold-trained PureGIN predictors. CCI coefficients are exact1234 = 0.30, dropout015 = 0.25, lowfusion = 0.10, aug050 = 0.20 and exact2025 = 0.15. Outer coefficients are CCI = 0.30, GIN2026 = 0.24, GIN3407 = 0.18 and GIN777 = 0.28. Both levels were inherited from the random-split model and held fixed for this evaluation.

The recorded CCI configuration initializes the molecular language model from pretrained ChemBERTa, uses the scaffold training/validation inputs, fits the target transformation on scaffold train, and selects checkpoints by scaffold validation MAE. Scaffold PureGIN predictions are joined by sample identifier. The final scalar prediction is the fixed weighted sum of component columns supplied in S4_scaffold_predictions.csv.

Table S17. Scaffold metrics recomputed from saved predictions.

Subset	n	MAE	RMSE	MedAE	MAPE (%)	R2
val	7035	30.3995	56.5302	16.6848	3.6202	0.8927
test	7857	35.1168	66.7544	19.0301	3.9599	0.8676

Full member arguments and the fixed-coefficient protocol are supplied in configs/scaffold_cci_arguments.json and configs/scaffold_protocol.json. The recorded server runtime is listed in Table S12.

S4.1. Target-system PredRet adaptation protocol

For each chromatographic system, ten outer folds were evaluated using an inner-training subset and inner-validation subset drawn from the outer-training portion. Preprocessing fits, checkpoint selection, early stopping and ensemble-weight selection used these inner subsets. Held-out outer-test records were evaluated with the selected predictors and fixed fold-specific coefficients.

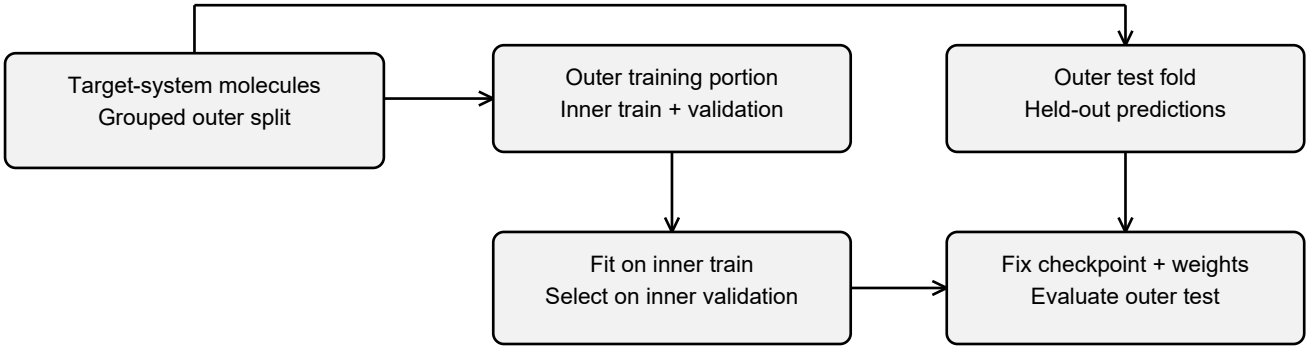


Figure S1. Evaluation flow for each target-system outer fold. Model selection uses inner validation; outer-test labels supply the final evaluation.

Table S18. Predictor construction in target-adapted MCRL.

Predictor	Initialization and target-system fitting
Four CCI members	exact1234, exact777, exact2025 and div_dropout015_seed777; initialize from corresponding SMRT-supervised checkpoints; update all model parameters on target-system inner train
Three PureGIN members	Seeds 2026, 3407 and 777; new graph model trained on each target-system inner-training subset
Descriptor predictor	Twelve RDKit descriptors; arithmetic mean of HistGradientBoosting and ExtraTrees predictions fitted on inner train
Eight-candidate fusion	Simplex coefficients selected by inner-validation MAE; uniform weights, individual predictors and 8,000 Dirichlet samples; seed 20260909
Local ChemBERTa-BiLSTM baseline	Initialized from B_chemberta_bilstm_seed2026; target-system adaptation on the same nested partitions

S4.2. Target-system optimization settings

Table S19. PredRet adaptation and selection settings.

Item	Setting
CCI optimizer	AdamW; encoder learning rate 0.000005; other parameters 0.00005; weight decay 0.001
CCI batches / selection	Train batch 8; evaluation batch 16; maximum 40 epochs; patience 6; minimum validation MAE
CCI augmentation / loss	Dropout 0.15; SMILES augmentation probability 0.20; power target transform; MAE task loss
CCI RTCL parameters	Temperature 0.10; omega_align 0.01; omega_fusion 0.005; similarity weight 0.10; ranking ratio objective
PureGIN	Train batch 128; maximum 180 epochs; patience 25; seeds 2026, 3407 and 777
HistGradientBoosting	StandardScaler; max_iter 350; learning_rate 0.04; l2_regularization 0.05
ExtraTrees	500 trees; minimum samples per leaf 2; random seed 20260909
Baseline	ChemBERTa layers 2/4/6 + BiLSTM; first-token pooling; maximum 256 tokens; batches 16/32; maximum 200 epochs; patience 20; learning rates 0.0001/0.00001; weight decay 0.001

The twelve descriptor features are molecular weight, MolLogP, TPSA, HBD, HBA, rotatable bonds, ring count, aromatic ring count, fraction Csp3, atom count, heavy atom count and bond count. The scaler in the HistGradientBoosting pipeline is fitted on inner-training descriptors.

The baseline source checkpoint is the local ChemBERTa-BiLSTM run with fixed-SMRT test MAE 32.7605 s. Its initialization record and target adaptation arguments are supplied in configs/predret_baseline_metadata_example.json.

For each system, mean and SD are calculated from the ten outer-fold metric values. Pooled MAE across all out-of-fold observations is retained as a separately named CSV field. Delta MAE is adapted-MCRL fold-mean MAE minus baseline fold-mean MAE.

S4.3. Ten-system PredRet results

Table S20. Outer-fold MAE, mean +/- sample SD (s).

System	n	Adapted MCRL	Local baseline	Delta MAE
Cao_HILIC	520	88.32 +/- 19.61	94.66 +/- 19.03	-6.34
Eawag_XBridgeC18	364	68.08 +/- 9.08	71.81 +/- 12.55	-3.73
FEM_Lipids	86	120.55 +/- 43.04	129.76 +/- 53.49	-9.21
FEM_long	412	142.75 +/- 35.30	156.79 +/- 34.08	-14.04
FEM_short	136	154.75 +/- 31.67	135.47 +/- 22.95	19.27
IPB_Halle	76	53.37 +/- 28.92	47.60 +/- 32.68	5.76
LIFE_new	173	26.05 +/- 8.43	25.04 +/- 5.93	1.01
LIFE_old	183	15.76 +/- 3.38	18.96 +/- 5.13	-3.21
MTBLS87	146	131.27 +/- 47.62	143.68 +/- 49.02	-12.41
UniToyama_Atlantis	101	131.43 +/- 43.19	127.64 +/- 33.07	3.79

Table S21. Outer-fold RMSE (s) and mean R2.

System	Adapted RMSE	Baseline RMSE	Adapted R2	Baseline R2
Cao_HILIC	142.79 +/- 26.27	150.40 +/- 25.51	0.5941	0.5549
Eawag_XBridgeC18	90.99 +/- 12.58	95.91 +/- 15.25	0.8385	0.8212
FEM_Lipids	160.82 +/- 64.82	182.89 +/- 74.28	0.7310	0.6629
FEM_long	239.17 +/- 108.84	277.80 +/- 120.83	0.9371	0.9172
FEM_short	220.71 +/- 55.04	179.11 +/- 41.52	0.6480	0.7473
IPB_Halle	78.22 +/- 38.53	66.32 +/- 46.23	0.6686	0.7766
LIFE_new	39.19 +/- 11.99	40.98 +/- 10.18	0.7782	0.7716
LIFE_old	23.46 +/- 7.41	29.54 +/- 9.14	0.8490	0.7722
MTBLS87	189.23 +/- 78.64	205.38 +/- 79.75	0.3942	0.2848
UniToyama_Atlantis	174.02 +/- 56.22	161.59 +/- 46.10	0.6315	0.6646

The system-results CSV also supplies MedAE, MAPE, metric SD values and pooled MAE. All 2,197 out-of-fold prediction pairs are included in S4_predret_outer_test_predictions.csv.

S4.4. Fold-level results (1-25 of 100)

Fold identifiers are zero-based. Train and validation refer to inner subsets; test is the outer fold. MAE values use the same cleaned RT labels for both methods.

Table S22. Per-fold sample counts and outer-test MAE (s).

System	Fold	Train	Val	Test	Adapted	Baseline	Delta
Cao_HILIC	0	421	47	52	89.689	84.758	4.931
Cao_HILIC	1	421	47	52	68.297	78.065	-9.768
Cao_HILIC	2	421	47	52	77.201	81.620	-4.419
Cao_HILIC	3	421	47	52	87.453	90.433	-2.980
Cao_HILIC	4	421	47	52	122.067	131.378	-9.311
Cao_HILIC	5	421	47	52	101.469	118.208	-16.739
Cao_HILIC	6	421	47	52	116.293	113.049	3.243
Cao_HILIC	7	421	47	52	74.787	78.525	-3.738
Cao_HILIC	8	421	47	52	82.612	88.527	-5.916
Cao_HILIC	9	421	47	52	63.314	81.994	-18.681
Eawag_XBridgeC18	0	294	33	37	57.772	47.116	10.656
Eawag_XBridgeC18	1	294	33	37	66.562	70.262	-3.700
Eawag_XBridgeC18	2	294	33	37	67.706	75.569	-7.863
Eawag_XBridgeC18	3	294	33	37	83.661	78.183	5.478
Eawag_XBridgeC18	4	295	33	36	61.376	69.028	-7.652
Eawag_XBridgeC18	5	295	33	36	81.660	91.015	-9.355
Eawag_XBridgeC18	6	295	33	36	58.425	69.919	-11.494
Eawag_XBridgeC18	7	295	33	36	73.569	86.862	-13.294
Eawag_XBridgeC18	8	295	33	36	61.958	59.912	2.046
Eawag_XBridgeC18	9	295	33	36	68.142	70.227	-2.085
FEM_Lipids	0	69	8	9	82.371	78.630	3.741
FEM_Lipids	1	69	8	9	73.058	97.333	-24.275
FEM_Lipids	2	69	8	9	181.806	204.769	-22.963
FEM_Lipids	3	69	8	9	128.648	127.631	1.017
FEM_Lipids	4	69	8	9	150.492	154.749	-4.257

Eight predictor coefficients per fold are supplied in S4_predret_fold_weights.csv. Per-fold RMSE, MedAE, MAPE and R2 are supplied in S4_predret_fold_results.csv.

S4.4. Fold-level results (26-50 of 100)

Fold identifiers are zero-based. Train and validation refer to inner subsets; test is the outer fold. MAE values use the same cleaned RT labels for both methods.

Table S23. Per-fold sample counts and outer-test MAE (s).

System	Fold	Train	Val	Test	Adapted	Baseline	Delta
FEM_Lipids	5	69	8	9	91.772	96.038	-4.266
FEM_Lipids	6	70	8	8	152.907	149.025	3.883
FEM_Lipids	7	70	8	8	179.230	227.398	-48.168
FEM_Lipids	8	70	8	8	87.354	85.236	2.117
FEM_Lipids	9	70	8	8	77.871	76.790	1.081
FEM_long	0	333	37	42	160.365	170.511	-10.146
FEM_long	1	333	37	42	113.245	125.472	-12.227
FEM_long	2	334	37	41	121.895	160.116	-38.221
FEM_long	3	334	37	41	192.069	189.821	2.247
FEM_long	4	334	37	41	84.979	94.214	-9.235
FEM_long	5	334	37	41	143.491	198.516	-55.025
FEM_long	6	334	37	41	145.201	139.514	5.687
FEM_long	7	334	37	41	196.720	184.585	12.135
FEM_long	8	334	37	41	114.985	125.921	-10.936
FEM_long	9	334	37	41	154.509	179.221	-24.712
FEM_short	0	110	12	14	182.367	115.327	67.040
FEM_short	1	110	12	14	143.252	139.889	3.363
FEM_short	2	110	12	14	132.282	155.147	-22.866
FEM_short	3	110	12	14	200.209	119.522	80.687
FEM_short	4	110	12	14	153.818	109.777	44.041
FEM_short	5	110	12	14	95.131	119.308	-24.178
FEM_short	6	111	12	13	195.633	155.644	39.989
FEM_short	7	111	12	13	152.240	122.514	29.725
FEM_short	8	111	12	13	136.668	135.484	1.183
FEM_short	9	111	12	13	155.875	182.111	-26.236

Eight predictor coefficients per fold are supplied in S4_predret_fold_weights.csv. Per-fold RMSE, MedAE, MAPE and R2 are supplied in S4_predret_fold_results.csv.

S4.4. Fold-level results (51-75 of 100)

Fold identifiers are zero-based. Train and validation refer to inner subsets; test is the outer fold. MAE values use the same cleaned RT labels for both methods.

Table S24. Per-fold sample counts and outer-test MAE (s).

System	Fold	Train	Val	Test	Adapted	Baseline	Delta
IPB_Halle	0	61	7	8	43.937	27.994	15.943
IPB_Halle	1	61	7	8	57.077	37.832	19.244
IPB_Halle	2	61	7	8	29.532	33.600	-4.068
IPB_Halle	3	61	7	8	48.353	28.758	19.596
IPB_Halle	4	61	7	8	26.195	33.756	-7.561
IPB_Halle	5	61	7	8	33.489	53.890	-20.401
IPB_Halle	6	62	7	7	46.194	20.757	25.437
IPB_Halle	7	62	7	7	116.545	86.618	29.928
IPB_Halle	8	62	7	7	91.940	123.578	-31.639
IPB_Halle	9	62	7	7	40.436	29.267	11.169
LIFE_new	0	139	16	18	38.823	34.281	4.542
LIFE_new	1	139	16	18	18.805	21.421	-2.615
LIFE_new	2	139	16	18	39.876	25.778	14.098
LIFE_new	3	140	16	17	25.166	20.922	4.243
LIFE_new	4	140	16	17	23.032	21.758	1.273
LIFE_new	5	140	16	17	25.737	24.892	0.845
LIFE_new	6	140	16	17	20.085	19.048	1.038
LIFE_new	7	140	16	17	32.614	36.541	-3.928
LIFE_new	8	140	16	17	21.380	25.395	-4.015
LIFE_new	9	140	16	17	15.006	20.378	-5.372
LIFE_old	0	148	16	19	21.869	17.228	4.641
LIFE_old	1	148	16	19	16.268	13.752	2.516
LIFE_old	2	148	16	19	15.575	22.381	-6.806
LIFE_old	3	149	16	18	10.701	19.757	-9.057
LIFE_old	4	149	16	18	19.108	19.950	-0.842

Eight predictor coefficients per fold are supplied in S4_predret_fold_weights.csv. Per-fold RMSE, MedAE, MAPE and R2 are supplied in S4_predret_fold_results.csv.

S4.4. Fold-level results (76-100 of 100)

Fold identifiers are zero-based. Train and validation refer to inner subsets; test is the outer fold. MAE values use the same cleaned RT labels for both methods.

Table S25. Per-fold sample counts and outer-test MAE (s).

System	Fold	Train	Val	Test	Adapted	Baseline	Delta
LIFE_old	5	149	16	18	15.832	14.104	1.728
LIFE_old	6	149	16	18	14.165	30.638	-16.473
LIFE_old	7	149	16	18	11.146	13.742	-2.595
LIFE_old	8	149	16	18	17.742	17.262	0.480
LIFE_old	9	149	16	18	15.174	20.817	-5.643
MTBLS87	0	118	13	15	141.288	147.140	-5.852
MTBLS87	1	118	13	15	71.227	74.492	-3.264
MTBLS87	2	118	13	15	113.950	125.499	-11.549
MTBLS87	3	118	13	15	171.936	171.628	0.308
MTBLS87	4	118	13	15	83.479	74.571	8.908
MTBLS87	5	118	13	15	164.515	189.314	-24.800
MTBLS87	6	119	13	14	146.953	163.650	-16.698
MTBLS87	7	119	13	14	85.537	108.914	-23.377
MTBLS87	8	119	13	14	223.849	228.287	-4.438
MTBLS87	9	119	13	14	109.935	153.271	-43.336
UniToyama_Atlantis	0	81	9	11	118.354	129.765	-11.410
UniToyama_Atlantis	1	82	9	10	84.450	120.566	-36.116
UniToyama_Atlantis	2	82	9	10	120.185	104.748	15.437
UniToyama_Atlantis	3	82	9	10	240.765	104.830	135.935
UniToyama_Atlantis	4	82	9	10	156.259	175.185	-18.926
UniToyama_Atlantis	5	82	9	10	119.318	161.065	-41.747
UniToyama_Atlantis	6	82	9	10	120.624	122.662	-2.038
UniToyama_Atlantis	7	82	9	10	108.917	117.225	-8.308
UniToyama_Atlantis	8	82	9	10	102.810	69.551	33.259
UniToyama_Atlantis	9	82	9	10	142.584	170.781	-28.198

Eight predictor coefficients per fold are supplied in S4_predret_fold_weights.csv. Per-fold RMSE, MedAE, MAPE and R2 are supplied in S4_predret_fold_results.csv.

S5. Companion data and configuration files

The archive preserves machine-readable values without rounding. CSV files use UTF-8 encoding and a header row. JSON files provide hierarchical configurations and protocol metadata. Source/configuration paths are project-relative. SHA-256 checksums are listed in package_manifest.json.

Table S26. Companion CSV tables.

File	Rows
S1_smrt_random_assignments.csv	77980
S1_smrt_scaffold_assignments.csv	77980
S1_smrt_split_statistics.csv	6
S1_predret_cleaned.csv	2197
S1_predret_system_statistics.csv	10
S1_overlap_checks.csv	66
S1_predret_smrt_shared_identities.csv	5
S1_predret_fold_assignments.csv	21970
S1_predret_nested_fold_checks.csv	100
S2_cci_member_configuration.csv	6
S2_puregin_member_configuration.csv	3
S2_puregin_metrics.csv	6
S2_smrt_ensemble_metrics.csv	4
S2_smrt_final_predictions.csv	14817
S3_baseline_metrics_and_sources.csv	3
S3_baseline_test_predictions.csv	23394
S3_ablation_run_metrics.csv	10
S3_ablation_test_predictions.csv	77980
S3_rtcl_seed_pairs.csv	3
S4_scaffold_metrics.csv	2
S4_scaffold_predictions.csv	14892
S4_predret_fold_results.csv	100
S4_predret_system_results.csv	10
S4_predret_fold_weights.csv	800
S4_predret_outer_test_predictions.csv	2197

S5.1. Configuration JSON and references

Table S27. Configuration and metadata inventory.

File	Contents
configs/cci_checkpoint_arguments.json	Recorded arguments and hashes for six CCI candidates
configs/puregin_checkpoint_arguments.json	Saved arguments, resolved architecture and hashes for three PureGIN members
configs/fusion_and_selection.json	Fusion coefficients, selection and target transformation
configs/ablation_run_arguments.json	Retained component and RTCL run arguments
configs/scaffold_protocol.json	Fixed-coefficient scaffold protocol
configs/scaffold_cci_arguments.json	Five active scaffold CCI configurations
configs/predret_adaptation_protocol.json	Target-system predictor construction and nested selection settings
configs/predret_baseline_metadata_example.json	Baseline initialization and adaptation metadata
metadata/data_protocol.json	Canonicalization, cleaning balance and fold protocol
metadata/recorded_environments.json	Runtime versions organized by operation
metadata/data_dictionary.json	Field names, descriptions, units and join keys
metadata/source_inventory.json	Source paths, byte sizes and SHA-256 hashes
metadata/validation.json	Completed consistency checks

References

S1. METLIN Small Molecule Retention Time (SMRT) resource. Figshare. DOI: 10.6084/m9.figshare.8038913. Original resource paper: Nature Communications, DOI: 10.1038/s41467-019-13680-7.

S2. PredRet: https://predret.org/. Original resource paper: Analytical Chemistry, DOI: 10.1021/acs.analchem.5b02287.

S3. SMILES-Morgan-HSCL source code: https://github.com/momomomo118/SMILES-Morgan-HSCL. Source snapshot: release source-20261019.